

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech CSE

Semester: III

Name of the Course: Python Programming

Course Code: TCS346(E)

Time: 3 Hours

Maximum Marks: 100

Note:

i. All questions are compulsory.

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	CO1 & CO2
(a)	Explain the concept of functions in Python and discuss the different types of arguments in Python functions with suitable examples. Develop a Python program to simulate a bank withdrawal process using exception handling. Assume an initial balance of ₹10,000. The program should accept a withdrawal amount and use all blocks—try, except, else, and finally. In the try block, check if the input is a numeric value, positive, and does not exceed the balance. Use separate except blocks for ValueError (non-numeric input), AssertionError (invalid or negative amount), and a general exception for unexpected errors. In the else block, deduct the amount and display the updated balance. The finally block should display "Thank you for banking with ABC Bank!"	
(b)	A travel agency named <i>WanderWorld</i> is creating a review formatter to standardize customer feedback before publishing on their website. Customers often type reviews inconsistently with random cases, symbols, and repeated spaces. The raw review string is: " travel with WANDERworld is fun and adventure!!! "	
(c)	Develop a Python program using string handling methods to: i. Remove extra spaces from both ends. ii. Convert the entire string to lowercase. iii. Replace "wanderworld" with "WanderWorld Travels". iv. Count how many times the word "and" appears. v. Check whether the cleaned review starts with the word "travel" and ends with "adventure". vi. Split the string into individual words and display them as a list.	

Q2	CO4																												
(a)	Develop a Python program to simulate a training management system for a company named <i>EduTech Solutions</i> using multiple inheritance. Create three classes — <i>Person</i> (with name and age), <i>Course</i> (with course_name and duration), and <i>Trainer</i> (which inherits from both <i>Person</i> and <i>Course</i>). The <i>Trainer</i> class should also have experience (in years) and a method to display all details. Use constructors (<code>__init__</code>) and the <code>super()</code> function where needed. Create one trainer object and display all the information.																												
(b)	Explain the concept of classes and objects in Python with suitable examples. Discuss how a class acts as a blueprint for creating objects, and how attributes and methods define an object's state and behavior. Describe the role of the <code>__init__()</code> constructor, the <code>self</code> keyword, and how multiple objects can be created and used within a program.																												
(c)	Explain the concept of polymorphism in Python with suitable examples. Describe how polymorphism allows the same function or method name to behave differently based on the object or the number of arguments. Discuss its implementation through method overriding in inheritance.																												
Q3	CO5																												
(a)	A college wants to analyze student performance using Python Pandas. The dataset <i>students.csv</i> (given below) contains student details with some missing values. <table border="1"><thead><tr><th>ID</th><th>Name</th><th>Marks</th><th>Attendance (%)</th></tr></thead><tbody><tr><td>1</td><td>Ramesh</td><td>85</td><td>92</td></tr><tr><td>2</td><td>Ananya</td><td>78</td><td>NaN</td></tr><tr><td>3</td><td>Suresh</td><td>NaN</td><td>88</td></tr><tr><td>4</td><td>Meena</td><td>91</td><td>95</td></tr><tr><td>5</td><td>Kavya</td><td>67</td><td>72</td></tr><tr><td>6</td><td>Arijun</td><td>NaN</td><td>NaN</td></tr></tbody></table>	ID	Name	Marks	Attendance (%)	1	Ramesh	85	92	2	Ananya	78	NaN	3	Suresh	NaN	88	4	Meena	91	95	5	Kavya	67	72	6	Arijun	NaN	NaN
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(b)	Explain the role of the Pandas library in data analysis and handling. Describe common DataFrame operations such as indexing, slicing, merging, and grouping. A college wants to visualize the average marks of students in five subjects using a bar chart in Python. The dataset is as follows: Subject=['Math', 'Physics', 'Chemistry', 'Biology', 'English'] Marks=[88,75,83,91,78] Develop a Python program using Matplotlib to plot a bar chart and answer the following sub-questions: - Plot the basic bar chart using the subjects as the x-axis and marks as the y-axis. - Change the color of the bars and add edgcolor and linewidth to make the bars visually distinct. - Use tick_label to display subject names under each bar and set width = 0.5 and align = 'center'. - Add alpha (transparency = 0.7) and a hatch pattern ('/' or 'x') to the bars for better visual appeal. - Add an appropriate label for the dataset, display a legend, and set zorder so that the bars appear on top of the grid lines.
Q4	CO6
(a)	Explain the concept of NumPy arrays and discuss how they differ from Python lists. Describe the process of array creation, indexing, and slicing with suitable examples. How do these features make NumPy more efficient for numerical computations compared to traditional Python data structures?
(b)	A retail company collected customer transaction data from multiple stores across different cities. However, the dataset contains several issues, such as missing values, duplicate entries, inconsistent date formats, and outliers in sales amounts. As a data analyst, describe in detail the data cleaning steps you would perform before analysis. Discuss suitable techniques such as handling missing data (mean, median, mode imputation), removing duplicates, data type correction, and outlier detection. Explain how these steps improve the accuracy and reliability of subsequent analysis.
(c)	Explain the concept of Regular Expressions (Regex) in Python and their importance in text processing. Discuss commonly used metacharacters and functions such as findall(), search(), match(), split(), and sub() from the re module with suitable examples.

Q5

CO5
&
CO3

Explain the Z-Score method for detecting outliers in a dataset. How does the Z-score help identify data points that deviate significantly from the mean?

Consider the following scenario:

A fitness center named FitLife records the daily calorie intake and hours of exercise for a group of members. The management suspects that a few data points might be outliers due to incorrect entries.

The dataset (fitlife.csv) is as follows:

Member_ID	Calories_Intake	Exercise_Hours
1	2200	1.5
2	2500	2
3	2300	1.8
4	5000	0.5
5	2600	2.2
6	2400	2
7	2550	1.9
8	2700	2.3
9	3000	3
10	8000	0.8

(a)

Implement a Python program using NumPy and Pandas to:

- Load the above dataset into a DataFrame.
- Calculate the Z-score for both `Calories_Intake` and `Exercise_Hours`.
- Identify and display rows where the absolute Z-score > 3 (considered as outliers).
- Interpret the results by listing which members' records appear to be outliers.

Explain the process of reading from and writing to files in Python. Differentiate between text files and binary files in terms of data storage, readability, and use cases. Describe the use of built-in functions such as `open()`, `read()`, `readline()`, `write()`, and `close()` with appropriate examples.

(b)

A bookstore named ReadMore wants to manage its inventory using Python data structures. Each book has details such as `Book_ID`, `Title`, `Author`, and `Price`. Write a short note on how the store can use lists and dictionaries to organize and manage this data. Explain how a list can be used to store multiple book records and demonstrate operations like indexing, slicing, and updating elements. Also describe how a dictionary can be used to map each `Book_ID` to its details, and how keys and values can be accessed, added, or modified. Discuss why these two data structures are suitable for developing a simple and efficient book management system.

(c)